SCG-HMH and data centres: transistor physics on the v8 numbers

Supplement to the SCG-HMH Closed Derivative Chain, v8 (V10 release package)

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New in V9, derived here, not from prior records. The authors' records imply this link without writing it out: the module runs on compute-hall waste heat, the v8 parasitic budget includes an AI-oracle microcontroller, and the amplifier register lists an AI heat-grid oracle (#32). This supplement works out the transistor side from public device physics and v8's own booked numbers. It adds no parasitic load to SCG-HMH and changes no booked number. Every figure below is either a v8 number, a physical constant, a value quoted from a cited source, or a derivation shown in full. Nothing here has been demonstrated on an SCG-HMH bench.

1. What the records already say (implied)

- v8 abstract: the module "runs on outside heat: the waste heat of systems already paid for (a compute hall, an industrial manifold)".
- v8 Section 3: outside heat "comes from a compute hall or waste-heat stream on Earth … The operator does not buy this heat."
- v8 change note 6: data-centre heat "is low-grade, and at an assumed 60 °C outlet it can return at most about 10 percent".
- v8 Table 27: the 0.8 kW module parasitic budget includes "the AI-oracle microcontroller and the Marx". The AI oracle at 2–10 % of P returns about 95 % as low-grade heat [A1 #32].
- Campus chilled-water co-function note (3 Sep 2026): data-centre heat is an accepted vaporiser inlet. That note "does not assert … a PUE number for any named data hall". This supplement keeps to the same rule.

2. Four ways transistor physics meets the generator

2.1 (a) The generator as a transistor-like amplifier (analogy)

A transistor lets a small control signal gate a large flow of power. SCG-HMH works the same way. A small electrical input arms the ring: the 0.8 kW module budget, inside which the Marx seeding is 1.0 W. The ring then delivers 99.55 MW, with the energy coming from free waste heat. The ratio is v8's own COP_e = 99.55 MW / 0.8 kW = 124,437, the "gain" in this picture. In a transistor, the power supply provides the energy and the gate only steers it. In SCG-HMH the waste heat provides the energy and the input electricity steers it. That is why the thermodynamic account COP_th = 0.2175 stays below 1. This is an analogy for explanation only. It is not a claim of power gain without an energy source.

2.2 (b) Chip heat back into power: the compute loop

Switching power in CMOS is $P_{dyn} = \alpha \cdot C \cdot V^2 \cdot f$, plus static leakage; [2] gives the scaling rules behind it. In practice nearly all the electricity a data hall draws ends up as heat. [5] tracks how much electricity data centres use worldwide. Here is the loop, derived for this supplement:

- E = the hall's electricity, all of which becomes heat.

- f_rec = the fraction of that heat delivered to the vaporiser band (a parameter, not booked).
- Electricity returned = COP_th · f_rec · E.
- Grid draw per unit of compute: G/E = 1 − COP_th · f_rec.
- Effective power multiplier = 1 / (1 − COP_th · f_rec).

f_rec	Return at COP_th = 0.2175 (v8 booked baseline)	Multiplier	Return at v8's 60 °C cap (~0.10)	Multiplier
1.00	21.75 %	1.278	10 %	1.111
0.95	20.66 %	1.260	9.5 %	1.105
0.50	10.88 %	1.122	5 %	1.053

The v8 numbers set both ends of the range. The 0.2175 figure is the booked design point with the published 40–180 °C heat band. A hall rejecting heat at around 60 °C is capped by v8's own second-law note at about 10 %.

Effective PUE, defined on grid electricity:

PUE_grid = PUE · (1 − COP_th · f_rec)

Here PUE is the hall's own figure, and f_rec applies to all of the facility's heat. For example, a hall at PUE 1.2 (an illustrative parameter, not a measured hall) with f_rec = 0.95 gives 1.2 × 0.7934 = 0.952 at COP_th 0.2175, or 1.2 × 0.905 = 1.086 at the 60 °C cap. The energy per computation that the grid sees falls by the same factor (1 − COP_th · f_rec). The compute itself is unchanged.

2.3 (c) Cryogenic CMOS on the cold side

SCG-HMH already holds LN_2 at 77 K as sink, working fluid and HTS coolant. Public device physics at 77 K:

- **Subthreshold swing.** The thermal limit is S = (kT/q)·ln10. That is 59.5 mV/dec at 300 K and 15.3 mV/dec at 77 K. Measured devices sit above this limit. A 0.18 µm MOSFET designed for 77 K reached 27 mV/dec, with V_th of 0.21 V and a 1 V supply, against a 4 V supply for its 300 K parent design [8]. Beckers et al. characterise 28-nm bulk CMOS at 300, 77 and 4.2 K [6].
- **Supply voltage and CV^2f.** At fixed C and f, P_dyn scales as V^2. For the published 77 K versus 300 K design pair in [8] (1 V against 4 V), that is a factor of $(1/4)^2 = 1/16$ in dynamic power for that pair. This ratio belongs to that 1994 pair and is not a forecast for modern nodes, which already run near 1 V.
- **Leakage** falls steeply as T drops [6][7].
- **Mobility.** Maximum carrier mobility rose by factors of 1.25 to 4.5 between room temperature and 77 K in SOI CMOS [9].
- **Interconnect.** Copper resistivity falls with temperature (tabulated in [10]), lowering wire resistance.

The "free" cooling, stated honestly. A refrigerator that pulls 1 W out at 77 K and rejects it at 300 K needs at least (300 − 77)/77 = 2.90 W of work (the Carnot limit). That is normally what kills cryo-computing. In SCG-HMH, the LN_2 already has to be boiled: the cycle needs heat put into the cold side to vaporise and expand the N_2. Chip heat at 77 K could do part of that vaporisation duty instead of dumping into a chiller. This supplement books nothing for it. The size of the duty, its effect on the tank inventory and recirculation, and any change to liquefaction load are plant questions for the bench. The cell's Phase-1 gate (σ > 5 S/m, net > 50 W) keeps priority, just as in the campus-chill note.

2.4 (d) The Landauer floor

Erasing one bit costs at least kT·ln2 [3], a limit verified experimentally [4]:

- At 300 K: 2.871×10^{-21} J (17.9 meV)
- At 77 K: 7.369×10^{-22} J (4.60 meV)
- Ratio: 300/77 = 3.90× lower at 77 K

Today's logic runs many orders of magnitude above this floor, so for now the gain is in principle only. The practical savings at 77 K come from 2.3.

3. What this does not claim

- No hall has been measured. PUE 1.2 is illustrative, and f_rec is a parameter.
- No new SCG-HMH parasitic, output or COP is booked. The 99.55 MW, 150,938 rpm, 5 T, COP_th 0.2175, COP_e 124,437 and train 9.9549 GW figures are unchanged.
- The cryo-CMOS numbers belong to the cited devices.
- The bench order is unchanged: spin to baseline rpm and measure σ first.

4. Bench and pilot steps (proposed)

1. Meter a real hall's heat-rejection temperature and flow, to find f_rec and which side of the 0.2175 / 0.10 range applies.
2. After the Phase-1 gate, run a small LN_2-immersed CMOS board as a measured vaporiser load, logging boil-off, tank level and σ.

References

[1] Smart, M. E.; Smart, J. K. SCG-HMH Closed Derivative Chain, v8 (this package; erratum of v7, https://doi.org/10.5281/zenodo.23291052). [2] Dennard, R. H. et al. Design of ion-implanted MOSFET's with very small physical dimensions. IEEE JSSC 9, 256 (1974). https://doi.org/10.1109/JSSC.1974.1050511 [3] Landauer, R. Irreversibility and Heat Generation in the Computing Process. IBM J. Res. Dev. 5, 183 (1961). https://doi.org/10.1147/rd.53.0183 [4] Bérut, A. et al. Experimental verification of Landauer's principle linking information and thermodynamics. Nature 483, 187 (2012). https://doi.org/10.1038/nature10872 [5] Masanet, E. et al. Recalibrating global data center energy-use estimates. Science 367, 984 (2020). https://doi.org/10.1126/science.aba3758 [6] Beckers, A.; Jazaeri, F.; Enz, C. Characterization and Modeling of 28-nm Bulk CMOS Technology Down to 4.2 K. IEEE JEDS 6, 1007 (2018). https://doi.org/10.1109/JEDS.2018.2817458 [7] Charbon, E. et al. Cryo-CMOS for quantum computing. IEDM 2016. https://doi.org/10.1109/IEDM.2016.7838410 [8] Yokoyama, M. et al. Short-channel-effect free 0.18 µm MOSFET by temperature-dimension combination scaling theory: design and experiment. IEEE EDL 15, 202 (1994). https://doi.org/10.1109/55.286692 [9] Young, K. K.; Tsaur, B.-Y. Operation of SOI CMOS devices at liquid-nitrogen temperature. IEEE EDL 11, 126 (1990). https://doi.org/10.1109/55.46955 [10] Matula, R. A. Electrical resistivity of copper, gold, palladium, and silver. J. Phys. Chem. Ref. Data 8, 1147 (1979). https://doi.org/10.1063/1.555614